HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and implemented **REST APIs** using **Java Spring Boot** to facilitate cross-service data exchange, incorporating request validation and circuit-breaker patterns.
- Developed and deployed **AI-powered applications**, integrating **Large Language Models (LLMs)** and **generative AI tools** into existing systems to automate business processes.
- Built automated workflows and **AI agents**, streamlining operational tasks by processing and responding to business requirements with scalable solutions.
- Implemented **Retrieval-Augmented Generation (RAG) architectures**, integrating **vector databases** and managing prompt engineering for context-aware responses.
- Collaborated with software development teams to embed intelligent capabilities into existing products, ensuring seamless integration of **AI functionalities**.
- Engineered **Python scripts** for data pipeline development and scheduled reporting, supporting ETL processes and data validation across enterprise systems.
- Containerized services using **Docker** and deployed them to a **Kubernetes cluster** (EKS), managing Helm chart configurations and resource allocation for predictable rollouts.

SKILLS

AI/ML Development – Large Language Models (LLMs), Generative AI, AI Agents, Retrieval-Augmented Generation (RAG), Vector Databases, Prompt Engineering, Machine Learning, Natural Language Processing (NLP)

Programming Languages – Python, Java, SQL

Frameworks & Libraries – Spring Boot, LangChain, LangGraph, LlamaIndex, Hugging Face, scikit-learn, pandas, NumPy, FastAPI

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Kubernetes, Jenkins, Git, CI/CD

Databases – PostgreSQL, MySQL, Oracle, Vector Databases

APIs & Integration – REST APIs, Third-Party AI Platforms

PROJECTS

AI Chatbot for Customer Support Automation

- Developed a conversational AI chatbot using **Python** and the **LangChain framework** to handle customer inquiries, reducing response times by approximately **40%** for common queries.
- Integrated the chatbot with a **vector database** to implement **Retrieval-Augmented Generation (RAG)**, enabling it to access and synthesize information from a knowledge base of enterprise documentation.
- Designed prompt templates and managed context to ensure accurate and relevant responses, improving user satisfaction scores by **15%** in pilot testing.

Intelligent Document Processing Pipeline

- Engineered a **Python-based pipeline** to ingest and process large volumes of unstructured documents, extracting key information using **NLP techniques**.
- Developed a proof-of-concept **machine learning model** for document classification, achieving **~0.88 AUC** on a dataset of 50,000 documents.
- Deployed the pipeline components as microservices on **AWS Lambda**, handling an average of **500 documents per hour**.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)